

## DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

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| Program: B. Tech. (Electronics and Computer Engineering)                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    | Semester: I   |    |                 |
| Course: Engineering Physics                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    | Code: ECBS102 |    |                 |
| Teaching Scheme (Hrs/week)                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        | Evaluation Scheme (Marks) |     |    |               |    |                 |
| Lecture                                                                                                                                                                                                                                                                                                                                                                                                  | Practical                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Tutorial | Credit | CIE                       | ETE | TW | OR            | PR | Total           |
| 02                                                                                                                                                                                                                                                                                                                                                                                                       | 02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -        | 03     | 40                        | 60  | 25 | -             | -  | 125             |
| Prerequisites:                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    |               |    |                 |
| Fundamentals of Physics, basic of interference, polarization, de-Broglie hypothesis, semiconductor and ultrasonic.                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    |               |    |                 |
| Course Objectives:                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    |               |    |                 |
| <div>1. To make the students understand and study the basic principles of Physics.</div> <div>2. To provide firm grounding to the students in the concept of physics to resolve many engineering and technological problems.</div> <div>3. To impart the knowledge of the fundamentals of physics to the students through hands on experiments and extend it to relevant engineering applications.</div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    |               |    |                 |
| Course Outcomes: After completion of this course, students will be able to -                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    |               |    |                 |
| CO1                                                                                                                                                                                                                                                                                                                                                                                                      | Explain basics of interference and polarization connected to engineering applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                           |     |    |               |    |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                      | Make use of Laser technology and Optical fiber in various disciplines.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                           |     |    |               |    |                 |
| CO3                                                                                                                                                                                                                                                                                                                                                                                                      | Outline the fundamentals of Quantum Physics and relate it to engineering applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                           |     |    |               |    |                 |
| CO4                                                                                                                                                                                                                                                                                                                                                                                                      | Apply basics of semiconductors for solving the engineering problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                           |     |    |               |    |                 |
| CO5                                                                                                                                                                                                                                                                                                                                                                                                      | Extend the understanding of Ultrasonic and NDT in engineering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |        |                           |     |    |               |    |                 |
| CO6                                                                                                                                                                                                                                                                                                                                                                                                      | Interpret the use of nanoparticles and superconductors in the field of engineering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |        |                           |     |    |               |    |                 |
| Course Contents:                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                           |     |    |               |    |                 |
| Unit                                                                                                                                                                                                                                                                                                                                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |        |                           |     |    |               |    | Duration (Hrs.) |
| 1.                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Wave Optics:</b><br><b>Units and its conversion</b> -Length, Mass, Velocity, Acceleration Momentum, Time, Temperature, Wavelength, Energy, Current, Voltage, Power, Intensity, Amplitude, Frequency, Pressure, Resistance, compressibility, resistivity, conductivity, Mobility, Angle.<br><b>Interference</b> - Interference in thin film of uniform thickness and its conditions (Simple Numerical), Engineering Applications – Ant-Reflection coating (ARC).<br><b>Polarization</b> - Polarization and its types, Malus law and Brewster’s law (Simple numerical), Double refraction, Huygens’s theory of double refraction, Differentiate between positive & negative crystal, Engineering applications of polarization: Liquid Crystal Display (LCD). |          |        |                           |     |    |               |    | 5               |
| 2.                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Laser and Optical Fiber:</b><br><b>Laser</b> - Basic Principles of laser, Elements of Laser, Characteristics of laser, He-Ne laser (Gas laser), Applications of laser – Medical, Industrial and Holography- Recording.<br><b>Optical fibers</b> - Propagation of light - Acceptance angle, Acceptance cone, Numerical aperture, Fractional Refractive Index Change (Simple numerical). Types of optical fibers, Advantages of optical fiber communication, Applications of optical fiber in Medical, Communication, Entertainment, Data Security.                                                                                                                                                                                                          |          |        |                           |     |    |               |    | 5               |

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| 3.           | <b>Quantum Physics:</b><br>de-Broglie hypothesis of matter waves, de-Broglie wavelength for a particle accelerated by Kinetic Energy (K.E) and a charged particle accelerated by Potential difference (PD) “V”, (Simple Numerical), Properties of matter waves, Heisenberg’s uncertainty principle for wide wave packet and narrow wave packet (Simple Numerical), Tunneling Effect, Engineering applications - Scanning Tunneling Microscope (STM), Introduction to Quantum Computing.                                                                                                                                                                                                                                          | 4         |
| 4.           | <b>Semiconductor Physics:</b><br>Classification of solids on the basis of band theory, Fermi level for metal and semiconductor, Position of Fermi level in extrinsic semiconductors (only diagram), Solar cell: principle, working, IV-characteristics, Efficiency and fill factor, Factor to improve efficiency of solar cell, Application, advantages and disadvantages of solar cell, Hall effect: derivation for Hall voltage and Hall coefficient (Simple numerical).                                                                                                                                                                                                                                                       | 5         |
| 5.           | <b>Ultrasonic and Non-destructive Testing:</b><br><b>Ultrasonic-</b> Properties of ultrasonic waves, Piezoelectric effect and inverse of piezoelectric effect, Generation of ultrasonic waves by inverse piezoelectric effect (using transistor), Compressibility of liquid by using ultrasonic waves (Simple Numerical).<br><b>Non- Destructive Testing (NDT):</b> Definition and its objectives, Difference between destructive testing and non-destructive testing, Application of NDT as an Ultrasonic flaw detection technique (Simple numerical), Advantages of NDT.                                                                                                                                                       | 4         |
| 6.           | <b>Nanophysics and Superconductivity:</b><br><b>Nanophysics-</b> Introduction of nanophysics, Properties of nanoparticles (Optical, Electrical, Mechanical), Applications of nanomaterials in Electronics, Automobile, Medical.<br><b>Superconductivity-</b> Definition of superconductivity on the basis of temperature dependence of resistivity, Properties of Superconductors, Meissner effect, Critical magnetic field (Simple Numerical), Type I and Type II Superconductors, Engineering applications of superconductivity in Superconducting Quantum Interface Device (SQUID) with its principle, working, general application of superconductors - Power Transmission, electronics, medical, principle of Maglev train. | 5         |
| <b>TOTAL</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>28</b> |

### List of Experiments:

#### Perform any (08) experiment out of 12:

- Experiment based on Newton’s rings (determination of wavelength of monochromatic light, determine radius of curvature of Plano-convex lens).
- Experiment based on polarization (To verify Law of Malus).
- Determination of refractive index using Brewster’s law.
- Experiment based on Double Refraction (Determination of refractive indices / Identification of types of crystal).
- Experiment based on Laser (Determination of thickness of wire / Number of lines on grating surface).
- Determination of Planck’s constant using available experimental setup.
- To study IV characteristics of Solar Cell and determine parameters (fill factor and efficiency).

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8. To determine Hall coefficient and charge carrier density.
9. Determination of velocity of ultrasonic waves and compressibility of given liquid by using Ultrasonic Interferometer.
10. An experiment based on optical fiber. (To determine the numerical aperture acceptance angle acceptance cone of optical fiber of laser diode.
11. Experiment based on semiconductor (To determine the temperature dependence characteristics of semiconductor).
12. To determine the unknown wavelength by using plane diffraction grating.
13. Study visit to research laboratory/ facility and submit report (**Compulsory**).

### Text Books:

1. M. N. Avadhanulu and P.G. Kshirsagar, "Engineering Physics", S. Chand Publications.
2. S. O. Pillai, "Solid State Physics", New age International Publications.
3. J. J. Sakurai, "Modern Quantum Mechanics", Pearson Publication.
4. V K Mehta and Rohit Mehta, "Basic Electrical Engineering", S Chand Publications.
5. Robert L. Jaffe and Washington Tayler, "The Physics of Energy", Cambridge University Press".

### Reference Books:

1. H. D. Young and R. A. Freedman, "University Physics", Pearson Publication.
2. Resnick and Halliday, "Principles of Physics", John Wiley and Sons.
3. Jenkins and White, "Optics", Tata McGraw Hill.
4. Noson S. Yanofsky and Mirco A. Mannucci, "Quantum computing for computer scientists", Cambridge University Press

### E-Resources:

1. NPTEL Course:
  - a) NPTEL lecture based on interference of polarized light by IIT Roorkee - [https://youtu.be/e-4QK\\_JVsdU?si=gWIBt41dDgeABO8Y](https://youtu.be/e-4QK_JVsdU?si=gWIBt41dDgeABO8Y)
  - b) NPTEL lecture based on Introduction of Polarization by IIT Roorkee- <https://youtu.be/fIVlzKB4bBQ?si=meWFP5matsopCABi>
  - c) NPTEL lecture based on Malus Law by IIT Roorkee <https://youtu.be/iFG82I3nFA0?si=JCln6fJqGNw6ix5U>
  - d) NPTEL lecture based on Double Refraction by IIT Roorkee <https://youtu.be/Pt5wvYyguq0?si=4mowxORZQXGXNxmW>
  - e) NPTEL lecture based on Semiconductor Physics by IIT Roorkee - <https://youtu.be/q7VIITSysMs?si=62lAMoJ2tMHKRiDH>
  - f) NPTEL lecture based on Introduction to superconductivity <https://youtu.be/hGPA1g8fKug?si=FdYfJju6bf6u2zRe>
  - g) NPTEL lecture based on Meissner Effect- <https://youtu.be/EkNnxBakJMs?si=qRnSvPID2NTE4rf->
2. Feynman lecture series: <https://www.feynmanlectures.caltech.edu/>
3. Concepts of Modern Physics, Arthur Beiser: - [https://nitsri.ac.in/Department/PHYSICS/Beiser\\_Modern\\_Physics.pdf](https://nitsri.ac.in/Department/PHYSICS/Beiser_Modern_Physics.pdf)
4. Lectures by Walter Lewin: <https://www.youtube.com/channel/UCiEHVhv0SBMpP75JbzJShqw>

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| <ol style="list-style-type: none"><li>5. Quantum Mechanics Lecture Series by Prof. H.C.Verma -<br/><a href="https://www.youtube.com/watch?v=JFWuAQRZPjQ&amp;list=PLWweJWdB_GuISnGkAafMpzzDBvTHg02At">https://www.youtube.com/watch?v=JFWuAQRZPjQ&amp;list=PLWweJWdB_GuISnGkAafMpzzDBvTHg02At</a></li><li>6. Virtual Labs, Amrita University- <a href="https://vlab.amrita.edu/?sub=1&amp;brch=195">https://vlab.amrita.edu/?sub=1&amp;brch=195</a></li><li>7. Virtual Labs, IIT Kanpur- <a href="https://bop-iitk.vlabs.ac.in/exp/energy-band-gap/simulation.html">https://bop-iitk.vlabs.ac.in/exp/energy-band-gap/simulation.html</a></li></ol> |
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